

Module 4-GENERATION, TRANSMISSION AND DISTRIBUTION

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Generation

- **Conventional sources**

- A. Hydro electric power stations

- B. Thermal power stations

- C. Nuclear power stations

- D. Diesel power plant

- **Non conventional sources**

- A. Solar Energy

- B. Wind Energy

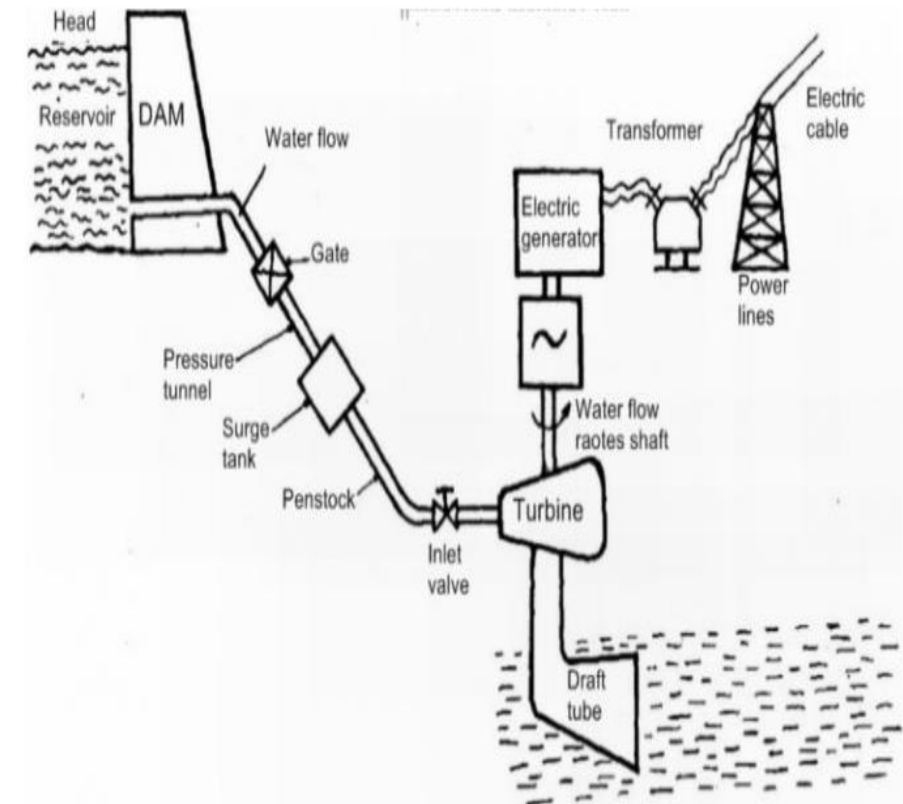
- C. Tidal Energy

HYDROELECTRIC POWERPLANT

A powerplant that produces by using flowing water to rotate a turbine which drives generator is called hydro electric power plant

- A reservoir is formed by constructing a dam across a flowing river. As more and more water collects in the reservoir, level of water behind the dam rises to a large height which has a lot of potential energy.
- A pressure tunnel is taken off from reservoir to gate/valve house at the start of pen stock. The gate or valve is provided for cutting off waterflow to power station and automatic isolating valves for cutting off water supply in case the pen stock bursts.
- A surge tank is also provided just before the valve house for better regulation of water pressure in the system.
- From the reservoir the water is carried to gate through pressure channel and from gate to turbine through pipes of large diameter made of steel or reinforced concrete called penstock.
- Water turbine converts hydraulic energy into mechanical energy and alternator coupled to water turbine converts mechanical energy into electrical energy.
- Water after doing useful work is discharged to tailrace.

Hydro-Power Plant (or Hydroelectric Power Plant)



Advantages and disadvantages of hydroelectric power plant

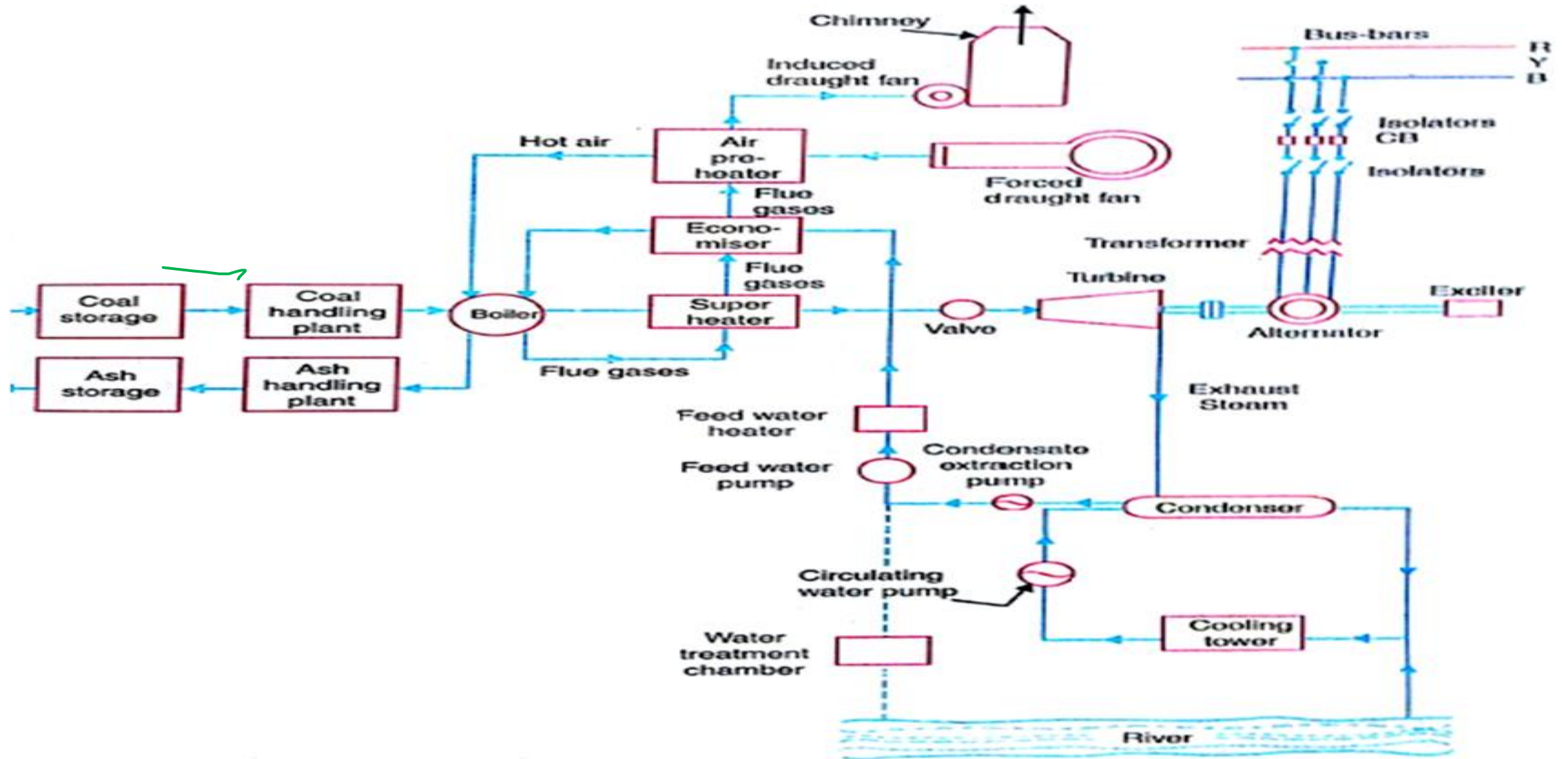
Advantages:-

- No fuel is required
- Reliable
- Cheaper
- Less maintenance
- Neat & clean plant because no smoke or ash is produced.
- Helps in irrigation and flood control.
- Highly skilled engineers are required only at the time of construction.

Disadvantages:-

- Requires large area.
- Construction cost is high.
- Long transmission lines are required as the plants are located in hilly areas.
- Depends on weather conditions.

Thermal Power Plant



PARTS OF A THERMAL POWER PLANT

1. Coal & Ash handling plant.
2. Steam generating plant.
 - a) Boiler
 - b) Super heater
 - c) Economizer
 - d) Air preheater
3. Steam Turbine.
4. Alternator / AC generator
5. Feed water
6. Cooling arrangement

PARTS OF A THERMAL POWER PLANT

1. Coal & Ash handling plant:-

Coal from storage is delivered to coal handling devices where it is crushed into small pieces. The crushed coal is fed to boiler and it is burned. Ash produced as a result of combustion of coal collects at the back of the boiler and is removed to ash storage through ash handling equipment.

2. Steam Generating plant

It consists of a boiler for the production of steam and other auxiliary equipment for the utilization of flue gases.

(a) Boiler:-

- The heat of combustion of coal is utilized to convert water into steam at high temperature and pressure.
- The flue gas from the boiler passes through superheater, economiser, airpreheater and are finally exhausted to atmosphere through chimney

(b) Superheater

- Steam produced in the boiler is wet and is passed through a superheater where it is dried and superheated by flue gases on their way to chimney
- The superheated steam is fed to steam turbine through main valve.

(c) Economiser

- An economiser is essentially a feedwater heater and derives heat from flue gases for this purpose.
- Feedwater is fed to economizer before supplying to boiler
- Economiser extracts a part of heat of flue gases to increase feedwater temperature.

(d) Airpreheater

- Air is drawn from the atmosphere by a forced draught fan and is passed through preheater before supplying to boiler.
- An airpreheater increases temperature of air supplied to boiler by deriving heat from flue gases.

3. Steam turbine

- The dry and superheated steam from the superheater is fed to the steam turbine through the main valve.
- The heat energy of steam when passing over the blades of turbine is converted to mechanical energy.
- After giving heat energy to the turbine, steam is exhausted to the condenser which condenses exhausted steam by means of cold water circulation

4. Alternator

- Steam turbine is coupled to an alternator. The alternator converts mechanical energy into electrical energy.

5. Feedwater

The condensate from the condenser is used as feedwater to the boiler. The water lost during the process may be suitably made up by supplying from external source

6. Cooling arrangement

- In order to improve efficiency of plant, steam exhausted from the turbine is condensed by condenser.
- Water is drawn from a natural source of water such as river, canal or lake and is circulated through condenser.
- Hot water from the condenser is passed on to the cooling tower where it is cooled.
- Cold water from cooling tower is reused as condenser.

Advantages and disadvantages of thermal power plant

Advantages

- Fuel used is cheaper.
- Less space in comparison with that of hydro-electric plants.
- Cheaper in production cost.
- Steam power plants can be installed at any place irrespective of the existence of fuels.

Disadvantages

- High maintenance and operating costs.
- Pollution of atmosphere due to fumes and residues.
- Requirement of water in large quantity.
- Disposal of ash is quite difficult.
- Long time for erection.

Nuclear Power Plant

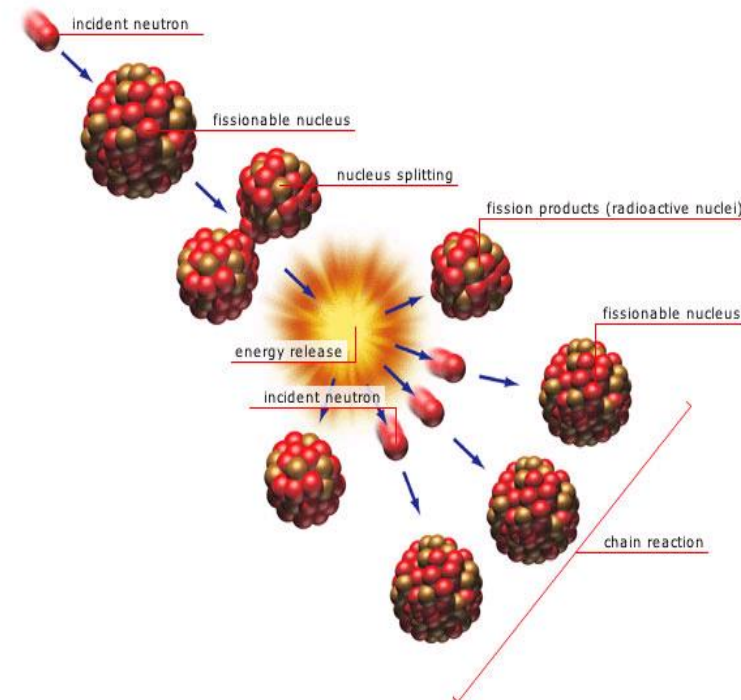
Nuclear Fission

➤ Splitting of heavier nuclei into lighter nuclei.

➤ The sum of the masses of the resulting nuclei is less than the original mass.

➤ The “missing mass” is converted to energy according to $E=mc^2$

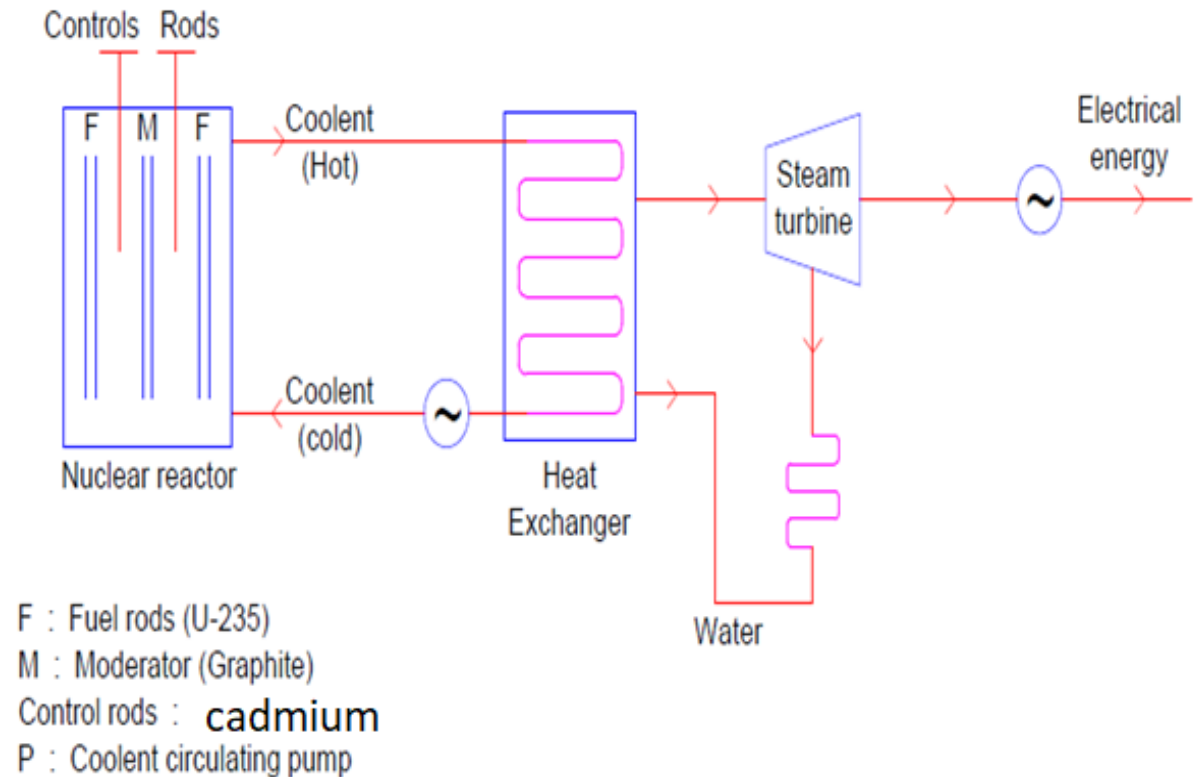
Neutron induced in U^{235}



BLOCK SCHEMATIC FOR NUCLEAR POWER PLANT

PARTS OF A NUCLEAR PLANT

- Nuclear reactor
- Heat exchanger
- Steam turbine
- Condenser
- Alternator



Schematic diagram of a Nuclear Power
Station indicating Main Components

Working of Nuclear PowerPlant

- **Nuclear Reactor** is an apparatus in which nuclear fuel is subjected to nuclear fission which controls the chain reaction.
- It houses fuel rods of uranium, moderator (graphite) and control rods(cadmium).
- **Fuel rods** consists of the fission material and releases large amount of energy when bombarded with slow moving neutrons.
- ✓ • **Moderator** generally graphite encloses the fuel rods which slows down the neutrons before they bombards the fuel rods
- ✓ • **Control rods** are of cadmium which absorb neutrons and regulate the supply of neutrons.
By adjusting the position of control rods, intensity of chain reaction can be increased or decreased.
- ✓ • Heat produced in the reactor is removed by a coolant, generally sodium metal. → Molten sodium
- **Heat Exchanger** Coolant gives up the heat to the heat exchanger which is utilized in raising the steam.
- **Steam Turbine** Steam produced in the heat exchanger is led to the steam turbine .After doing work, steam is exhausted to condenser. Condenser condenses the steam

Advantages and disadvantages of Nuclear power plant

Advantages

❖ Less quantity of fuel for generation of given amount of energy compared to other sources.

❖ High reliability, Efficient and less running cost.

Disadvantages

❖ Fuel is expensive and not available in abundance everywhere.

❖ High capital cost.

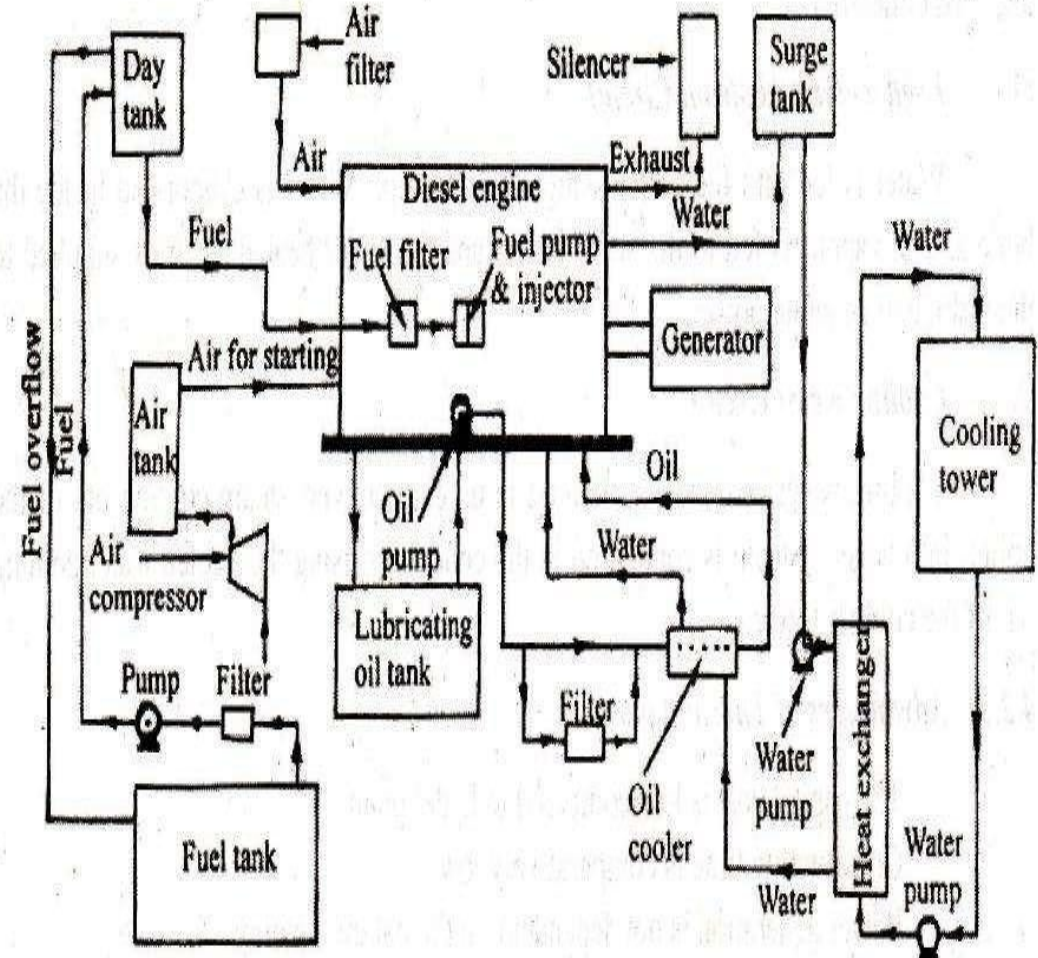
❖ Maintenance cost is very high.

❖ Nuclear waste disposal is a great problem

Diesel power plant

- Diesel Engine
- Air intake system
- Fuel supply system
- Exhaust system
- Cooling system
- Lubricating system
- Generator

When the diesel fuel burning inside the engine, its start to produce a mechanical power. The combustion of diesel fuel produces increased temperature & pressure inside the engine. Due to this pressure gases are formed, this gas pushes the piston inside the diesel engine, and then mechanical power is produced. With the use of this mechanical power the shaft of diesel engine starts rotating.



Diesel power plant

(i) Fuel supply system

Fuel from the storage tank is pumped through a filter into a smaller tank called all day tank. This tank supplies the daily requirements of the diesel engine.

(ii) Air intake system

The air required for the combustion of fuel inside the diesel engine cylinder is drawn through the air filter. The purpose of the filter is to remove dust from the incoming air.

(iii) Exhaust system

The exhaust gases coming out of the engine are very noisy. In order to reduce the noise a silencer is used.

(iv) Cooling system

- The temperature of the burning fuel inside the engine cylinder is 1500°C to 2000°C . In order to lower this temperature, water is circulated around the engine.
- The hot water leaving the jacket is passed through the heat exchanger.
- The heat from the heat exchanger is carried away by the raw water circulated through the heat exchanger and is cooled in the cooling tower.

(v) Lubricating system

- This circuit includes lubricating oil tank, oil pump and oil cooler.
- The purpose of the lubrication system is to reduce the wear of the engine moving parts. Parts of the cylinder such as piston, shafts, valves must be lubricated.
- Lubrication also helps to cool the engine.

Advantages and disadvantages of diesel power plant

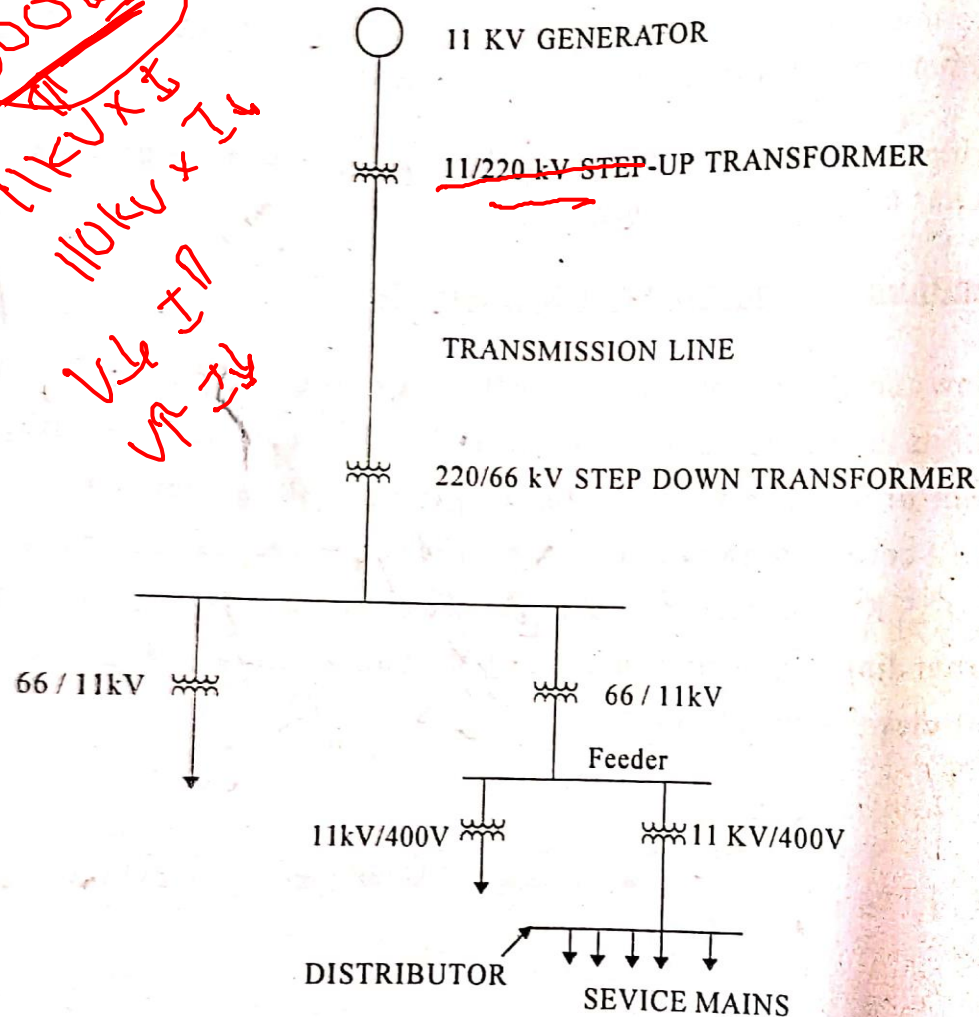
Advantages

1. Very simple design also simple installation and occupies less space.
2. Limited cooling water requirement.
3. Diesel power plant are more efficient than steam power in the range of 150MW capacity.
4. Quickly started and put on load.
5. It can respond to varying loads without any difficulty.
6. Smaller storage is needed for the fuel.
7. Layout of power plant is quite simple.
8. There is no problem of ash handling.
9. Less supervision required

Disadvantages

1. High Maintenance, lubrication cost and operating cost.
2. Fuel cost is more, since in India diesel is costly.
3. The plant cost per kW is comparatively more.
4. The life of diesel power plant is small due to high maintenance.
5. Noise is a serious problem in diesel power plant.
6. Diesel power plant cannot be constructed for large scale.

5.6 ONE LINE DIAGRAM OF A TYPICAL POWER SYSTEM



(i) Primary transmission:

The electric power at 220kV is transmitted by 3 phase 3 wire overhead system. This forms primary transmission.

(ii) Secondary transmission

In 220kV substation, the voltage is stepped down to 66kV by stepdown transformer. From here, the electric power at 66kV is transmitted by 3 phase 3 wire overhead system. This forms secondary transmission.

(iii) Primary distribution

In 66kV substation, the voltage is stepped down to 11kV by stepdown transformer. 11kV line run along important road sides of city. This forms primary distribution to industrial purposes

(iii) Secondary distribution

The electric power from primary distribution(11kV) is delivered to distribution transformer. Here voltage is reduced to 400V. From transformer, power is transmitted at 400V by 3 phase 4 wire over head system to the consumers.

Advantages & limitations of high transmission voltage

Advantages

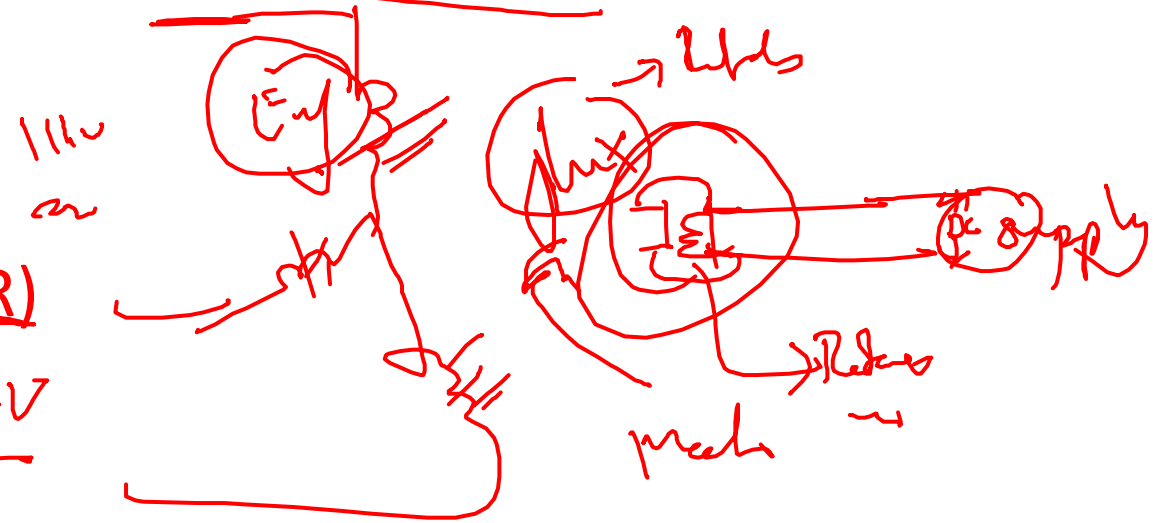
1. With increase in the transmission voltage size of the conductors is reduced (Cross section of the conductors reduce as current required to carry reduces).
2. Reduction in copper losses due to reduced current
3. Transmission efficiency is increased due to low line losses.
4. Due to low current, voltage drop will be less . This leads to better voltage regulation.

Limitations

- With the increase in the voltage of transmission, the insulation required between the conductors and the earthed tower increases. This increase the cost of line support
- With increase in the voltage of transmission, more clearance is required between conductors and ground. Hence higher towers are required.
- With increase in the voltage transmission, more distance is required between the conductors. Therefore cross arms should be long

Electrical equipment in power stations/Power plants

1. AC generator/Alternator
2. Exciter → DC supply
3. Automatic voltage regulator (AVR)
4. Power transformer 11 kV / 220 kV
5. Busbars
6. Switching equipment
7. Protective equipment
8. batteries



Electrical equipment in power stations/Power plants

(i) AC generator/Alternator

- It is a device which converts mechanical energy to electrical energy. A generator forces electric current to flow through an external circuit.
- Generator varies according to the type of power plant.
- ❖ Salient pole alternator normally used in hydro electric power plant.
- ❖ Non salient pole alternator is used in thermal power stations.
- Generator decides size of power plant

(ii) Exciter

- Excitation system is required to provide necessary field current(DC current) to the rotor winding of alternator(synchronous AC generator)

(iii) Automatic voltage regulator(AVR)

- Synchronous generators have inherently large internal reactance and, therefore, high voltage regulation. So, in order to maintain terminal voltage of the generator under various operating conditions, provision of automatic voltage regulation equipment is essential.

The main functions of an AVR are as follows:

1. Control of system voltage within prescribed limits and have the operation of the machine nearer to the steady state stability limit.
2. Proper division of reactive load between the alternators operating in parallel.
3. Prevention of dangerous over-voltages on the occurrence of sudden loss of load on the system.

Electrical equipment in power stations/Power plants

(iv) Power transformer

Power transformer are used for stepping-up the voltage for transmission at generating stations

(v) Busbars

- * Bus bar is a main bar or conductor carrying an electric current to which many connections are to be made.
- * Usually two buses are provided in a system one is called the “Main” bus and other “Auxiliary ” or transfer bus
- * There are many shapes of bus bar available like round tubes, round solid bars or square tube
- * Material used for bus bar is aluminum because it has high corrosion resistance than copper and lower cost



(vi) Switching equipment

During the operation of power system the generating plants, transmission lines, and other electrical equipment are required to be switched on and off under both normal and abnormal conditions Switchgears are used

(viii) Protective equipment

- * Protective equipment are very important in system to isolate the abnormal conditions
- * Feature of protective equipment
 1. greater reliability
 2. High speed of operation
 3. Simple and withstand to large value of fault
- * Various Protective equipment are Fuse, Circuit Breakers, Relays, Lightning arresters

Electrical equipment in power stations/Power plants

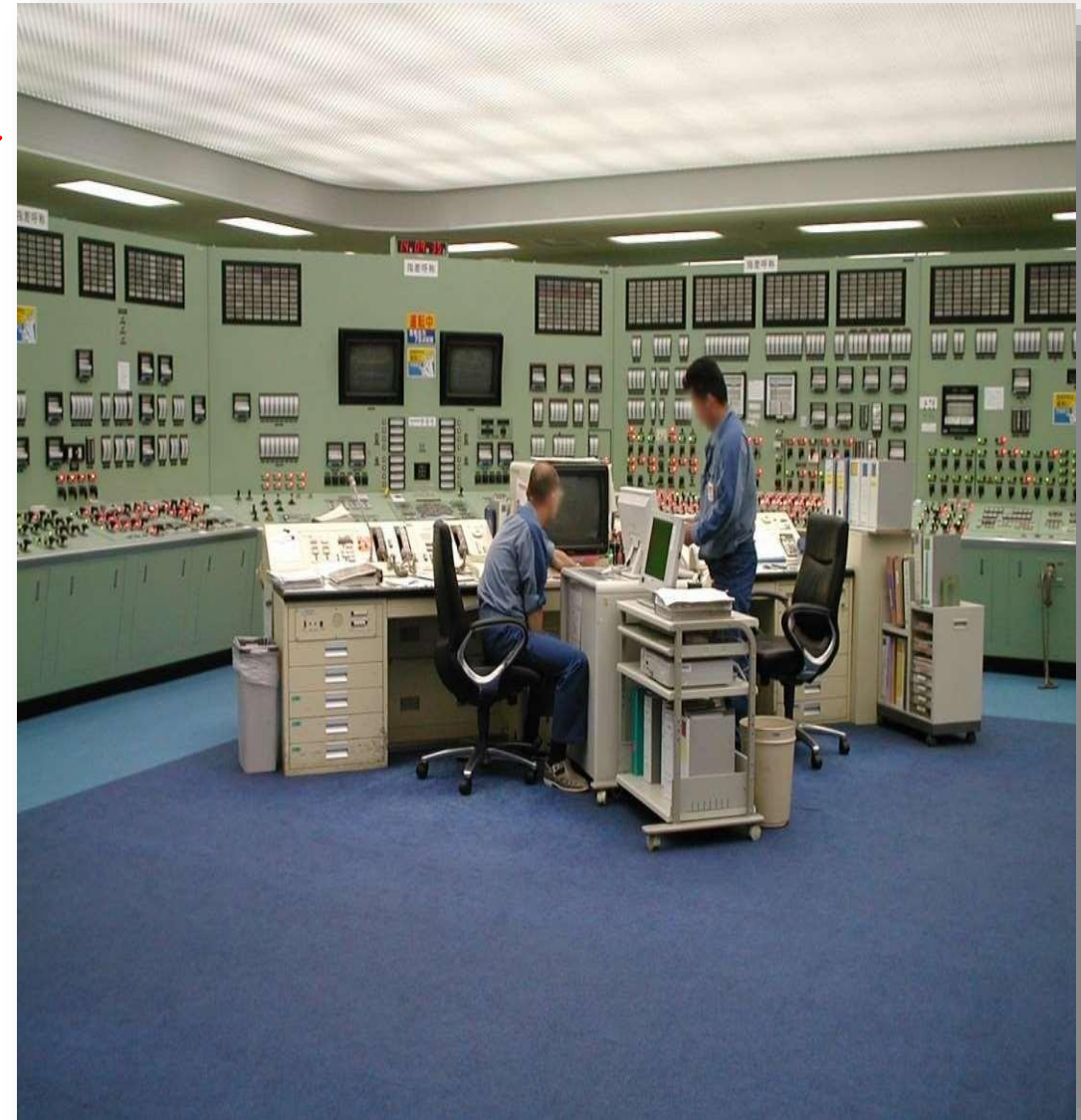
(ix) Batteries

- * All power plants, DC supply for protection and control purpose
- * DC supply is provided from Storage batteries.
- * Lead acid batteries mostly used in ~~power~~ stations because their higher cell voltage and lower cost



Control room

- * Control room is nerve centre of the power station
- * Various controls like Voltage adjustments, load control, emergency tripping of turbine and other equipment and instruments are housed in the control room.
- * Location of control room is away from noise and should be near to the switch house
- * There should be no glare, neat and clean, well ventilated and free from draughts



Methods of power transmission

1. DC system
2. Single phase AC system
3. Three phase AC system

Types of transmission line

1. Overhead line

- Overhead transmission line uses bare conductors.
- These conductors placed at a height from the ground. To maintain clearance between the conductors and ground, supporting towers are used.
- The voltage of the transmission line decides the height of the tower.
- The insulators used to provide insulation between the conductor and the tower.

2. Underground cables

- In the underground transmission system, the number of conductors bunched together with proper insulation.
- The underground cable provided with lead sheet and armoring. These provide protection against moisture and mechanical injury.
- As the voltage level increase, the thickness of insulation increases.

S.No.	Particular	Overhead system	Underground system
1.	Public safety	It is less safe.	It is more safe.
2.	Initial cost	It is less expensive.	It is more expensive.
3.	Faults	Faults occur frequently.	Very rare chances of faults.
4.	Appearance	It gives shabby look.	Its appearance is good as wires are not visible.
5.	Flexibility	It is more flexible as new conductors can be laid along the existing conductors.	It is not flexible, as new conductors are to be laid in new channels.
6.	Location of fault	Fault point can be easily located.	Fault point cannot be easily located.
7.	Repair	Can be easily repaired.	Cannot be easily repaired.
8.	Working voltage	It can work upto 400 kV.	It can work only upto 66 kV due to insulation difficulty.

Substation and classification of substation

Main function of substation are

- (i) To receive energy transmitted at high voltage from generating station
- (ii) To reduce the voltage to a value appropriate for local distribution
- (iii) To provide facilities for switching.
- (iv) They provide points where safety devices may be installed to disconnect equipment or circuit in the event of fault.



Classification

1. According to service requirement

(a) **Transformer substation:-** changes voltage level

(b) **Power factor correction substation:-**
Improves powerfactor

(c) **Converting substation**

Changes AC to DC power

2. According to constructional features

(a) **Indoor substation:-** substation equipments are installed indoor upto 11kV

(b) **Outdoor substation:-** For voltages beyond 66kV, equipments are installed outdoor.

(c) **Underground substation:-** equipments are placed underground. In thickly populated areas and price of land is high.

SUBSTATION EQUIPMENTS

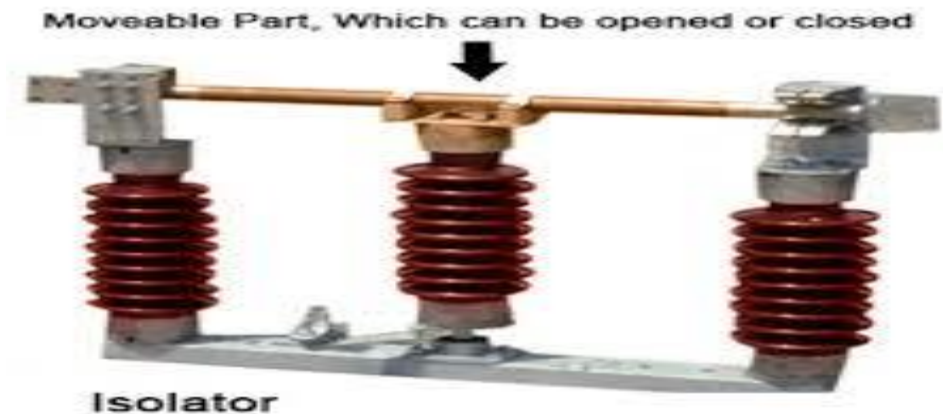
1. Power transformers

Power transformers are used in transmission network of higher voltages for step- up and step down application (400 kV, 200 kV, 110 kV, 66 kV, 33kV) and are generally rated above 200MVA



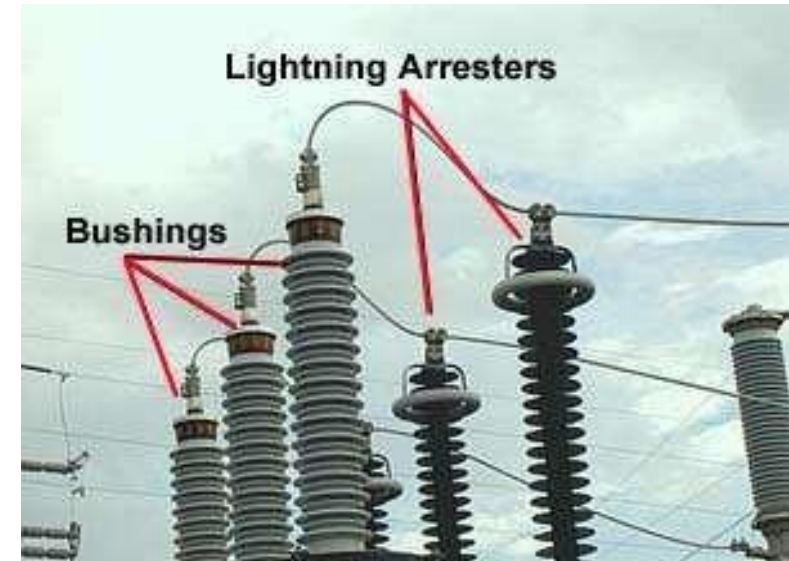
2. Isolator

- It is a disconnection switch and to be operated on no load manually.
- It is used to make sure that an electrical circuit can be completely de-energized for service or maintenance.
- It provide electrical isolation of the equipment .



3. Lightning Arrester

- It discharge the over voltage surges to earth and protect the equipment insulation from switching surges and lightning surges.
- It located at the starting of the substation as seen from incoming transmission lines and is the first equipment of the substation



4. Instrument transformers

The main purpose of an instrument transformer is to decrease high current as well as voltages for a secure & realistic value. These values can be calculated with conventional devices. This transformer is also used for triggering the protective relay (AC type) by providing the current as well as voltage. They are of

a. Potential transformer

Potential transformers are instrument transformers. They have a large number of primary turns and a few number of secondary turns.

It converts voltages from high to low so that meters can handle

a. Current transformer

- CT is a type of instrument transformer that is used in power system for measurement, detection, protection of the system.
- Current transformers are used extensively for measuring current and monitoring the operation of the power grid.



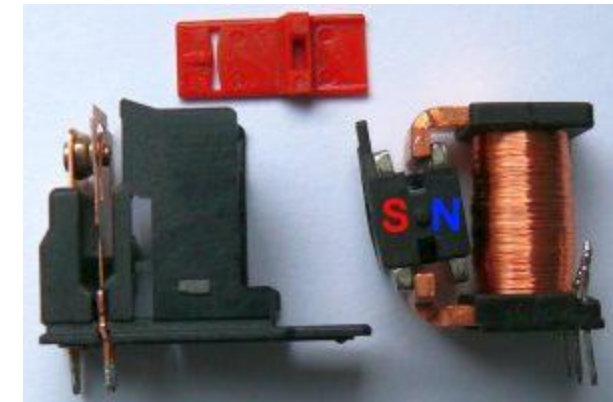
5. Circuit breaker

- It designed to protect an electrical circuit from damage caused by overload or short circuit.
- Its basic function is to detect a fault condition and discontinue the electric flow
- It includes two moving parts that are usually closed. When an error happens in the system, then the relay transmits the signal to the circuit-breaker & therefore their parts are moved separately.



6. Relays

The relay is an electrical device, and the main role of this device in the substation is, it guards the grid component against the irregular conditions like faults. This is one type of detecting device, used to detect and determine the fault location, and then it sends the signal to the circuit breaker. After receiving the signal from the relay, the circuit breaker will detach the faulted part.

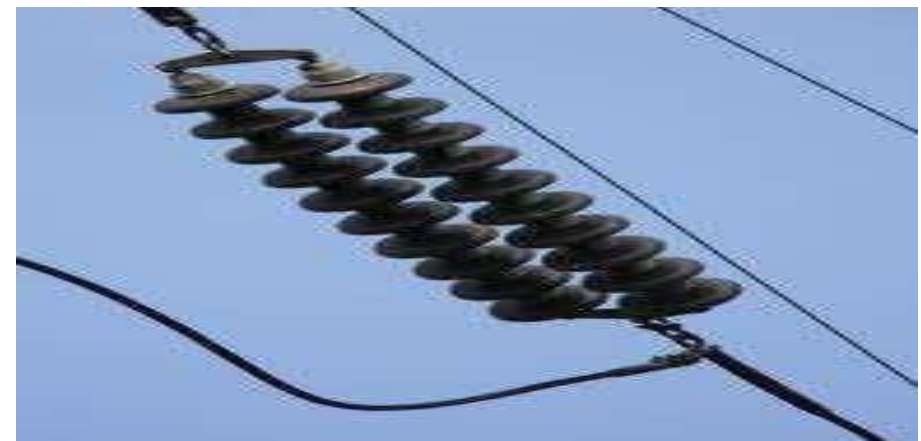


6. Busbar

The bus bar is a very important component in an electrical substation. It is a kind of current carrying conductor where many connections are made

7. Insulators

- An insulator, also called a dielectric, is a material that resists the flow of electric current.
- Insulators are used in electrical equipment to support and separate electrical conductors without allowing current through themselves



8. Batteries

In large power stations or substations, the operation of lighting, relay system, or control circuits are powered by batteries.

9. Capacitor bank

This device is inbuilt with capacitors that are connected either in series or else parallel. The main function of this is to store the electrical energy in electrical charge form. The capacitor bank is an efficient method to improve power factor



10 .Wavetrap

❑ It trap the high frequency communication signals sent on the line from the remote substation and diverting them to the telecom/teleprotection panel in the substation control room .

❑ The Line trap offers high impedance to the high frequency communication signals thus obstructs the flow of these signals in to the substation bus bars.

❑ The signals are primarily teleprotection signals and in addition, voice and data communication signals.



Metering, controlroom and relay panels



- To house various measuring Instruments, control Instruments, Protective relays. They are located in air-conditioned building. Control Cables are laid between Switchyard equipment and these panels.

Corona

- Corona Discharge (also known as the Corona Effect) is an electrical discharge caused by the ionization of a fluid such as air surrounding a conductor that is electrically charged.
- Corona discharge can cause an audible hissing or cracking noise as it ionizes the air around the conductors. This is common in high voltage electric power transmission lines. The corona effect can also produce a violet glow, production of ozone gas around the conductor, radio interference, and electrical power loss.



Explanation:-

- The corona effect occurs naturally due to the fact that air is not a perfect insulator – containing many free electrons and ions under normal conditions. When an electric field is established in the air between two conductors, the free ions and electrons in the air will experience a force. Due to this effect, the ions and free electrons get accelerated and moved in the opposite direction.
- The charged particles during their motion collide with one another and also with slow-moving uncharged molecules. Thus the number of charged particles increases rapidly. If the electric field is strong enough, a dielectric breakdown of air will occur and an arc will form between the conductors.

Factors Affecting Corona

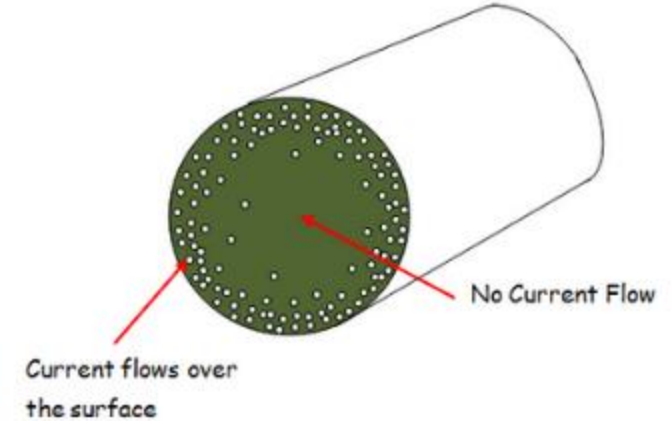
- **Atmosphere:** As it is already explained that the corona forms due to ionization of the air. There are always some free electrons in the air (which means air is pre-ionized to a little extent). However, in stormy weather, the number of free electrons is more than that in normal conditions. In such case, corona occurs at much lesser voltage.
- **Conductor size:** Corona discharge also depends on the shape and size of the conductors. Irregularities on the conductor surface concentrate the electric field at locations, resulting in corona at these spots. Thus, a stranded conductor gives rise to more corona than a solid conductor with a smooth surface. Also, [conductors having large diameter](#) have lower electric field gradient at the surface. Hence, conductors having large diameter produce lower corona than small-diameter conductors.
- **Spacing between the conductors:** Larger distance between the conductors reduces the electric stresses between them. And, hence, larger the distance between conductors, lesser the corona formation.
- **Line voltage:** As it is already explained, lesser the line voltage, lesser the ionization of surrounding air. Corona discharge starts to occur when the voltage becomes greater than a minimum critical voltage called as critical disruptive voltage.

How To Reduce The Corona Discharge?

- Corona discharge is always accompanied by power loss (which is dissipated in the form of sound, light, heat and chemical action). Though it accounts for a small percentage of total losses, power loss due to corona becomes significant in foul or wet weather conditions. Corona discharge can be reduced by the following methods:
 - **By increasing the conductor size:** As explained above, larger the diameter of the conductor, lesser the corona discharge.
 - **By increasing the distance between conductors:** Larger the conductor spacing, lesser the corona.
 - **Using bundled conductors:** Using a [bundled conductor](#) increases the effective diameter of the conductor. This results in reduction of the corona discharge.
 - **Using corona rings:** The electric field is greater where the conductor curvature is sharp. Therefore, corona discharge occurs first at the sharp points, edges and corners. To mitigate this, corona rings are employed at the terminals of very high voltage equipments such as at the bushings of a very high voltage [transformer](#) (Corona discharge also occurs in high voltage equipment). A corona ring is electrically connected to the high voltage conductor, encircling the points where corona discharge may occur. This significantly reduces the potential gradient at the surface of the conductor, as the ring distributes the charge across a wider area due to its smooth round shape.

Skin effect

- Skin effect is a tendency for alternating current ([AC](#)) to flow mostly near the outer surface of an electrical conductor, such as metal wire. The effect becomes more and more apparent as the frequency increases
- The distribution of current over the entire cross-section of the [conductor](#) is quite uniform in case of a DC system.



The AC resistance is more than the DC resistance because **the alternating current flowing through the conductor does not distribute uniformly and concentrates near the surface of the conductor** and thus the effective cross-section area decrease and the resistance increase. The AC resistance of the same conductor is always more than the DC resistance because of the skin effect

- Load dispatching

Frequency and voltage is to be maintained within limits in addition to ensuring reliability of power supply and for maintaining the frequency and voltage within limits it is essential to match the generation of active and reactive power with the load demand.

In large power systems, a central load dispatcher is necessary to assign loads to various stations and units in accordance with a predetermined schedule, modified from time to time as the actual load differs from predicted load or as emergencies arise owing to loss of generating units or tie-lines. The load dispatching may be by telephone, remote telemetering and signaling, or both.

Functions of Load dispatch centre

- It provides system security
- Islanding facility
- Optimum use of resources
- Load forecasting and demand estimation
- Maintain power quality of power supply which is transmitted through the regional grid.
- Regulating the system frequency
- Help to quickly restore the system after any disturbances
- Reliable operation of the transmission system
- Matching the power generation to the power demand
- Load shedding implementation
- Reactive power management and voltage control